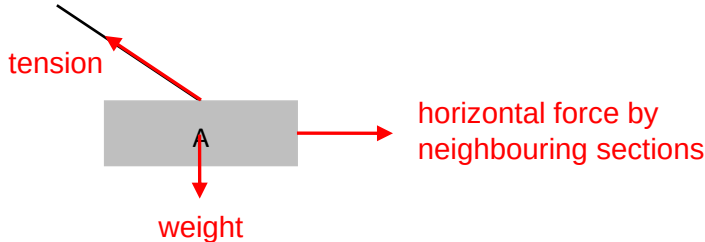


7	(a)	better to have many cables supporting the load instead of relying on one very strong pair of cables taking the same load	B2
	(b)	maximum mass support by each tower $= \frac{(1.4 + 8.5 + 11.5) \times 10^6}{2}$ C1 $= 1.07 \times 10^7$ A1	
	(c)	$\frac{(8.5 + 11.5) \times 10^6}{1600} \times 20$ C1 $= 2.5 \times 10^5 \text{ kg}$ A1	
	(d)	$\tan^{-1} \frac{8}{20}$ C1 $= 21.8^\circ$ A1	
	(e)	 tension (to the left and 21.8° above horizontal) (of cables on either side—left and right—of the roadway) (total) weight downwards horizontal force to the right (by neighbouring sections) it is in equilibrium with the resultant of the three forces being zero (award only if the forces drawn can balance)	B1 B1 B1 B1
	(f)	$\uparrow T \sin 21.8^\circ = 2.5 \times 10^5 \times 9.81$ $T = 6.60 \times 10^6 \text{ N}$ this T is due to 2 cables on either side of the roadway so tension in each cable = $\frac{T}{2} = 3.30 \times 10^6 \text{ N}$	C1 C1 A1

	(g)	$\text{increase in extension} = \frac{\text{increase in tension}}{\text{force constant}}$ $= \frac{5.2 \times 10^5}{7.0 \times 10^6}$ $= 0.074 \text{ m}$	C1 A1
	(h)	This is to ensure the cable is not stretched beyond its elastic limit which would result in the cable being permanently deformed. Or Any relevant safety	B1
	(i)	centre of the bridge because cables are pulling it in opposite directions	B1 B1